# CS & IT ENGINEERING


## Operating System

### Memory Management

Lecture -2


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# Recap of Previous Lecture

doubt ?

**Topic** Memory Management

**Topic** Contiguous Memory Management Technique

# Topics to be Covered

**Topic** Non-Contiguous Memory Management Technique

**Topic** Paging

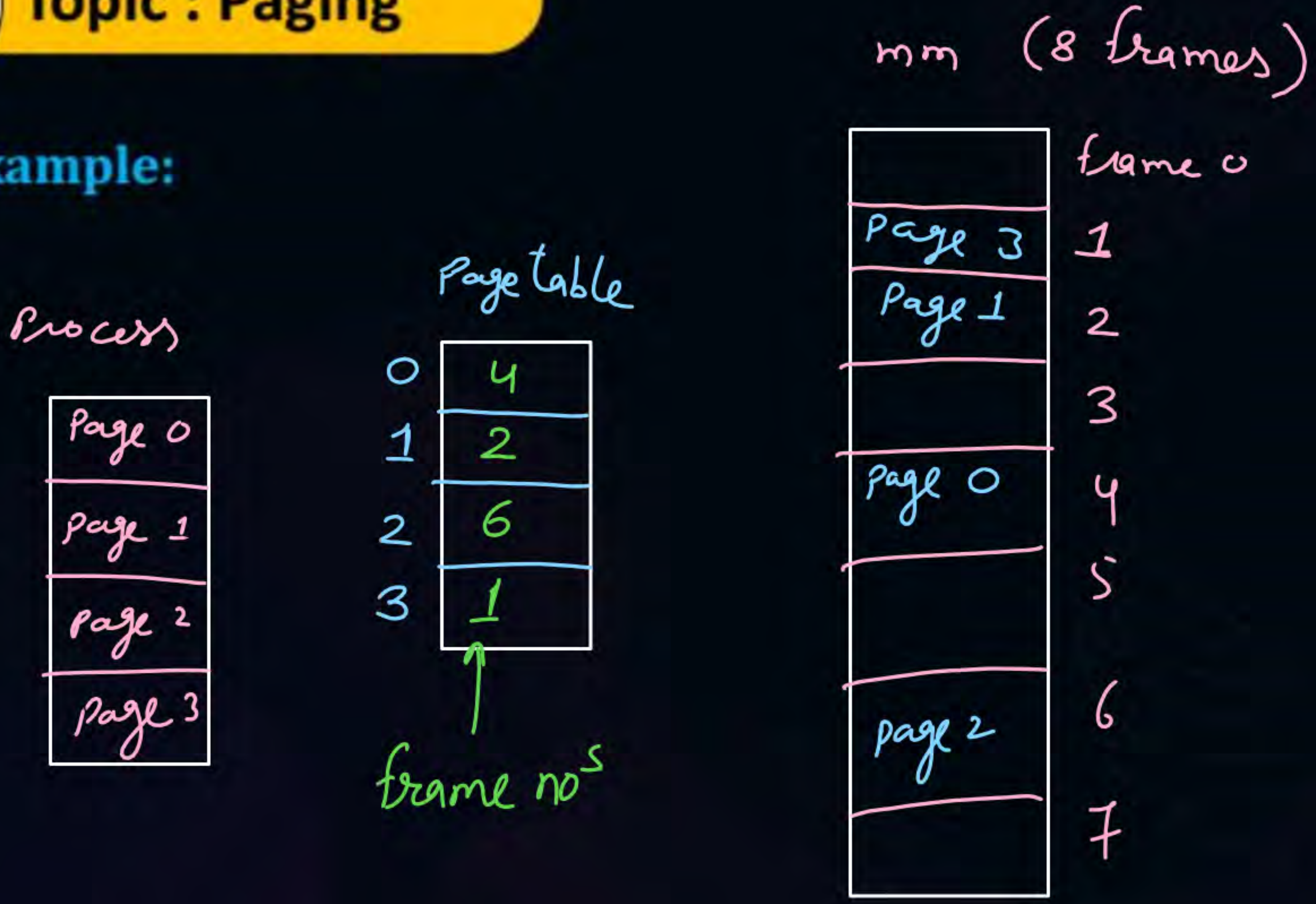
**Topic** Address Translation

# Topic : Paging

- Process is divided in equal size of partitions called as pages
- Physical memory is divided in same size of partitions called as frames
- Pages are scattered in frames
- A page table is maintains to map which page is stored in which frame.
- No. of entries in page table = no. of pages in process
- Page table size = No. of pages in process * 1 page table entry size
- 1 page table entry size = bits for frame no. + extra bits

# Topic : Paging

**Example:**



# Topic : Paging

Consider

- A process has 4 pages
- Main memory has 8 frames

Process

| | |
|---|---|
| 00 | Page 00 |
| 01 | Page 01 |
| 10 | Page 10 |
| 11 | Page 11 |

Page table

| | |
|---|---|
| 00 | 100 |
| 01 | 010 |
| 10 | 110 |
| 11 | 001 |

mm

| | |
|---|---|
| 000 | |
| 001 | Page 11 |
| 010 | Page 01 |
| 011 | |
| 100 | Page 00 |
| 101 | |
| 110 | page 10 |
| 111 | |

Assume page size = 2 bytes

process size = 4 * 2B = 8 bytes = $2^3$ B

mm size = 8 * 2B = 16 bytes = $2^4$ B $\Rightarrow$ address = 4 bits

0000

⋮

1111

Process size = Logical address space

mm size = physical address space

# Topic : Paging

physical address

logical address

**Process**

| Logical address | Content | Page |
|---|---|---|
| 000 | a | 00 |
| 001 | b | |
| 010 | c | 01 |
| 011 | d | |
| 100 | e | 10 |
| 101 | f | |
| 110 | g | 11 |
| 111 | h | |

**Page Table**

| Page | Frame |
|---|---|
| 00 | 100 |
| 01 | 010 |
| 10 | 110 |
| 11 | 001 |

**Physical Memory**

| Address | Content | Frame |
|---|---|---|
| 0000 | | frame 000 |
| 0001 | | |
| 0010 | g | 001 |
| 0011 | h | |
| 0100 | c | 010 |
| 0101 | d | |
| 0110 | | 011 |
| 0111 | | |
| 1000 | a | 100 |
| 1001 | b | |
| 1010 | | 101 |
| 1011 | | |
| 1100 | e | 110 |
| 1101 | f | |
| 1110 | | 111 |
| 1111 | | |

CPU generates logical address

This L.A. belongs to which page

L A

| Page no. | byte no within page |
|---|---|

ex:- LA = 100

| 10 | 0 |
|---|---|

page no = 10

byte no. = 0

search in P.T. => frame no. = 110

| frame no. | Byte no |
|---|---|
| 110 | 0 |

Physical add. = 1100

CPU generates L.A. $\boxed{011}$

$\boxed{01 \mid 1}$

page no. | byte no.

page no.

$\Downarrow$

search Page table

$\Downarrow$

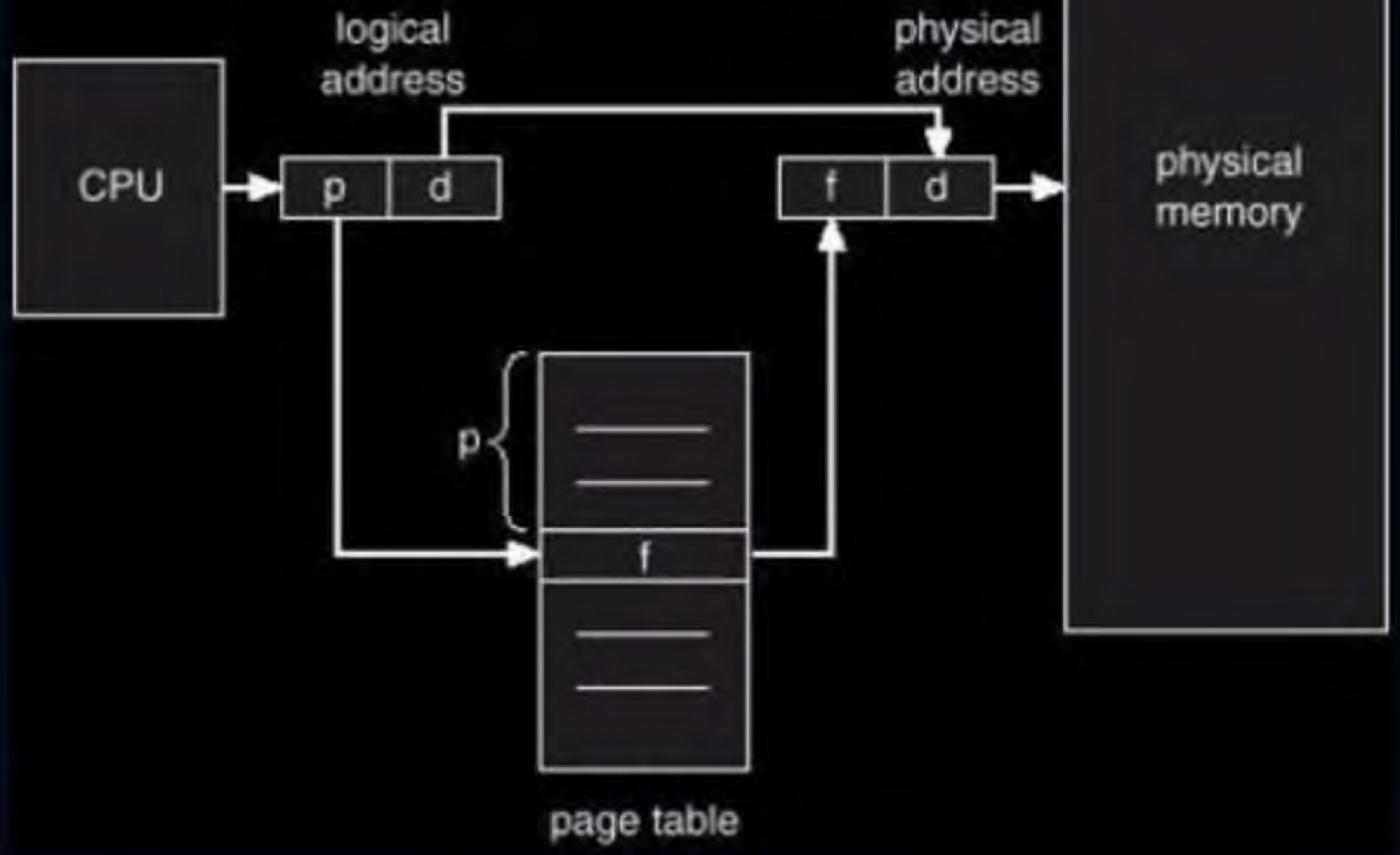
frame no. = 010

$\boxed{010 \mid 1} \Rightarrow \boxed{0101}$

physical address

# Topic : Paging

Logical address to physical address translation



## Topic : Paging

- Processor will have a view of process and its pages
- Page table is used to map a process page to a physical frame
- Number of entries in page table = Number of pages in process
- OS maintains a page table for each process ⇒ To provide mem. protection

L.A. is divided into 2 parts

| p | d |
|---|---|
| Page no. | byte no. |

P.A. is divided into 2 parts

| f | d |
|---|---|
| frame no. | byte no. |

no. of bits in d = $\log_2(\text{page size})$

no. of bits in p = $\log_2(\text{no. of pages})$

no. of bits in f = $\log_2(\text{no. of frames})$

page size = $2^{\text{bits in d}}$

no. of pages in process = $2^{\text{no. of bits for p}}$

no. of frames in mm = $2^{\text{bits for f}}$

| Page size | Byte no. in page (d) |
| --- | --- |
| 2 bytes | 0, 1 ⇒ 1 bit |
| 4 bytes | 00, 01, 10, 11 ⇒ 2 bits |
| 8 bytes | 000, .... 111 ⇒ 3 bits |
| $1KB = 2^{10} B$ | 10 bits |

| no of pages | bits for p |
| --- | --- |
| $4 = 2^2$ | 2 bits |
| $8 = 2^3$ | 3 bits |
| $16K = 2^{14}$ | 14 bits |
| $2^{17} = 128k$ | 17 bits |

$$\text{L.A. size} = \log_2(\text{L.A. space}) \text{ bits}$$

$$\text{P.A. size} = \log_2(\text{P.A. Space}) \text{ bits}$$

Ques) L.A. = 20 bits

P.A. = 24 bits

page size = 256 bytes = $2^8$ B $\Rightarrow$ d = 8 bits

L.A.

$\longleftarrow 20 \longrightarrow$

| p | d |
|---|---|
| 12 | 8 |

P.A.

$\longleftarrow 24 \longrightarrow$

| f | d |
|---|---|
| 16 | 8 |

page table size = no. of pages in process * 1 P.T. entry size

$$= 2^{12} * (16 + 0) \text{ bits}$$

$$= 2^{12} * 16 \text{ bits}$$

## Topic : Question

LA 29

| p | d |
|---|---|
| 19 | 10 |

PA 35

| f | d |
|---|---|
| 25 | 10 |

#Q. Consider a paged memory system where the logical address of 29 bits and physical address of 35 bits. If each page table entry is of 4 bytes and page size is 1KB then: $\rightarrow 2^{10}B \Rightarrow d = 10$ bits

4 bytes $\downarrow$ 32 bits

1. Number of pages in process? $2^{19}$
2. Number of frames in main memory? $2^{25}$
3. Number of bits for page number? 19 bits
4. Number of bits for frame number? 25 bits
5. Number of entries in page table? $2^{19}$
6. Page table size? $2^{19} * 4$ bytes

Ques) In previous Question no. of extra bits stored in each page table entry ___7___ bits?

Soln

$$1 \text{ P.T. entry} = \text{frame no.} + \text{extra bits}$$

$$32 \text{ bits} = 25 \text{ bits} + \text{extra bits}$$

$$\text{extra bits} = 7 \text{ bits}$$

## Topic : Question

[GATE-2001]

LA $\xleftrightarrow{32}$

| p | d |
|---|---|
| 20 | 12 |

PA $\xleftrightarrow{26}$

| f | d |
|---|---|
| 14 | 12 |

P.A = 26 bits ← $2^{26}$ B

logical

#Q. Consider a machine with 64 MB physical memory and a 32-bit virtual address space. If the page size is 4 KB, what is the approximate size of the page table?

A. 16 MB

B. 8 MB

C. 2 MB

D. 24 MB

size of P.T. = no. of pages in process * 1 P.T.E. size

$= 2^{20} * 14$ bits

= 14 M bits

$\simeq$ 2 MB

b ⇒ bits

B ⇒ bytes

H.W.

#Q. Consider a paged memory system where the process size is 16MB and main memory size is 4GB. The page size is 2KB.

1. Number of pages in process?
2. Number of frames in main memory?
3. Number of bits for page number?
4. Number of bits for frame number?
5. Number of entries in page table?
6. Page table size?

H.W.

#Q. Consider a paged memory system where the process size is 128MB and main memory size is 2GB. The page size is 1KB.

1. Number of pages in process?
2. Number of frames in main memory?
3. Number of bits for page number?
4. Number of bits for frame number?
5. Number of entries in page table?
6. Page table size?

H.W.

#Q. Consider a paged memory system where the logical address is 25 bits and physical address is 33 bits. The page size is 4KB.

1. Number of pages in process?
2. Number of frames in main memory?
3. Number of bits for page number?
4. Number of bits for frame number?
5. Number of entries in page table?
6. Page table size?

[MCQ] H.W. [GATE-2015]

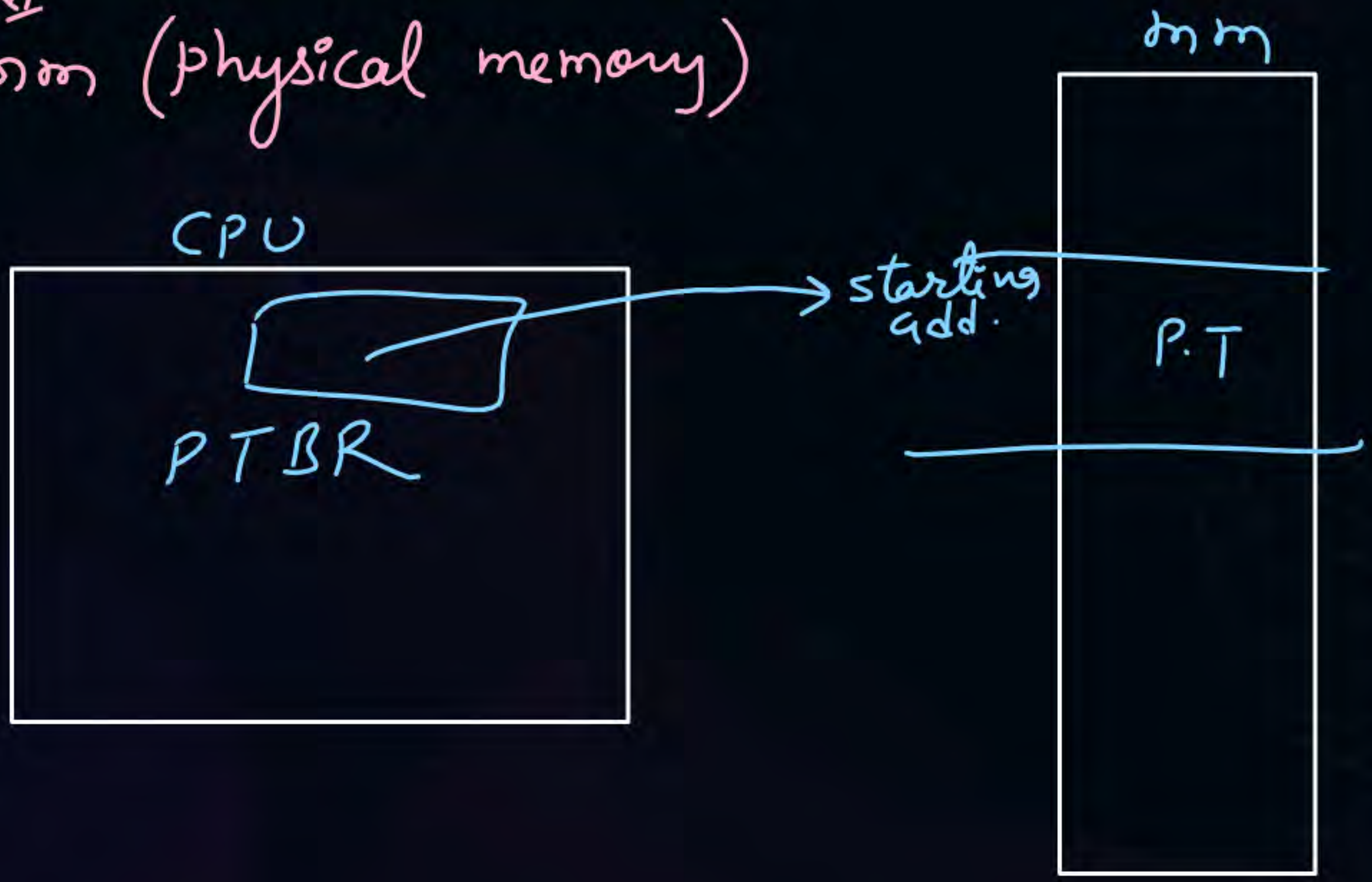
#Q. A computer system implements 8 kilobyte pages and a 32–bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the logical address supported by the system is _____ bits?

# Topic : Paging

## Where the Page table Stored?

$\Downarrow$

in mm (physical memory)




PTBR (Page Table Base Reg.)

$\Downarrow$

stores starting add. of P.T.

# Topic : Paging

## Performance of Paging

Effective mem. access time = $2 * t_{mm}$ $\begin{cases} \text{one for P.T.} \\ \text{one for content} \end{cases}$
(EMAT)

where

$t_{mm}$ = mm access time

if P.T. is very small and stored in CPU Register

$EMAT = t_{mm}$

because P.T. access time from Register will be negligible.

# 2 mins Summary

**Topic** Non-Contiguous Memory Management Technique

**Topic** Paging

**Topic** Address Translation

# Happy Learning

## THANK - YOU